RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra)
Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda
Designation / Role: Software & Data Engineer
Task Order Reference: 12 October 2026 – ETL Pipeline Assembly

2. DAILY LOG ENTRY FORM

- **Date:** 12 October 2026

- **1. Assigned Task:**

  ETL Pipeline Assembly — Generate -- Validate -- Process -- Store Pipeline Execution

- **2. What Was Studied:**

  Studied the ETL (Extract, Transform, Load) workflow and its application to synthetic sensor telemetry data. Examined dataset generation, CSV extraction, data-quality validation, record separation, data type conversion, timestamp processing and structured output storage. Also studied the use of Pandas Data Frames for tabular data manipulation and JSON for pipeline execution summaries.

- **3. What Was Implemented:**

  Developed and executed a standalone Python ETL pipeline in `Day8/codes/etl_pipeline.py`. Generated an independent synthetic sensor dataset containing 250 records, extracted the data using Pandas, and applied validation checks for timestamps, record identifiers, sensor IDs, flight phases and numeric readings. Separated accepted and rejected records, processed 242 accepted records, and saved the rejected records with validation reasons. Generated processed CSV data, rejected CSV data, a JSON execution summary and an execution log.

- **4.Tools / Libraries Used:**

  VS Code, PowerShell, Python 3.14.6, CSV ,Pandas , Pathlib , JSON.

- **5.Output File Name-** suraj_rudra_task/ Day8/codes/etl_pipeline.py
  Path-https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**
  No problem was encountered

- **7. How Problem Was Resolved:**
  No problem was encountered.

- **8. Next Task Scheduled:**

  (13 October 2026)- System auditing, structure JSON logging, and error trace capturing

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature: _____ Date: 12/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

[ ] Approved / Completed

[ ] Needs Revision / Delayed

[ ] Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026